

Problem Definition (1/2)

- We want to project our data X into another form Y which is less redundant (more informative).

Say we are given m data samples each of which has n features.

Given a dataset $X \in \mathbb{R}^{m \times n}$, find some orthonormal matrix P in $Y = PX$ such that $C_Y = \frac{1}{n} Y Y^T$ is a diagonal matrix.

Orthonormal matrix:

Matrices where the column vectors are orthonormal.

Orthonormal vectors:

Each column vector has length of one and is orthogonal to all the others.

Problem Definition (2/2)

- Why should C_Y be a diagonal matrix?
 - C_Y is a covariance matrix of Y .
 - If C_Y is diagonal, it means each feature of Y has zero covariance with the other features. 자기자신과의 분산만 존재하고 다른 성분과의 공분산은 0
 - That is, Y becomes to contain only orthogonal features (efficient representation)!
- Why should P be orthonormal?
 - If P is orthonormal, $P^T P = I$, meaning that P can only rotate or reflect X without changing its geometry. 모든 성분이 독립이고 NORM이 1 = 곱해지는 원본 데이터의 분산을 바꾸지 않음
 - If P is not orthonormal, the transformation can additionally scale or shear the data, artificially changing the variances and covariances rather than simply rotating the coordinate system.

Finding PCA using Eigenvector Decomposition (1/2)

- We begin with rewriting C_Y in terms of the unknown variable P :

$$\begin{aligned}C_Y &= \frac{1}{n} Y Y^\top \\&= \frac{1}{n} (P X) (P X)^\top \\&= \frac{1}{n} P X X^\top P^\top \\&= P \left(\frac{1}{n} X X^\top \right) P^\top \\&= P C_X P^\top\end{aligned}$$



$$\begin{aligned}C_Y &= P C_X P^\top \\&\text{C}_X\text{를 고유값 분해하면} \\&= P (E^\top D E) P^\top \\&= E (E^\top D E) E^\top \\&= D\end{aligned}$$

Remark1: D is a diagonal matrix, which is preferable for our purposes!

Remark2: By selecting P to be E , we can make C_Y to be diagonal!

전처리된 데이터 X 에 대해 X 의 공분산 행렬을 구한 다음 그 행렬을 고유값 분해하면 고유값 분해된 E 가 바로 PCA의 조건을 만족하는 P 가 됨

Finding PCA using Eigenvector Decomposition (2/2)

- Therefore, we can now determine what P should be.

$$\mathbf{C}_Y = \mathbf{P}\mathbf{C}_X\mathbf{P}^\top$$

$$= \mathbf{P}(\mathbf{E}^\top \mathbf{D} \mathbf{E})\mathbf{P}^\top$$

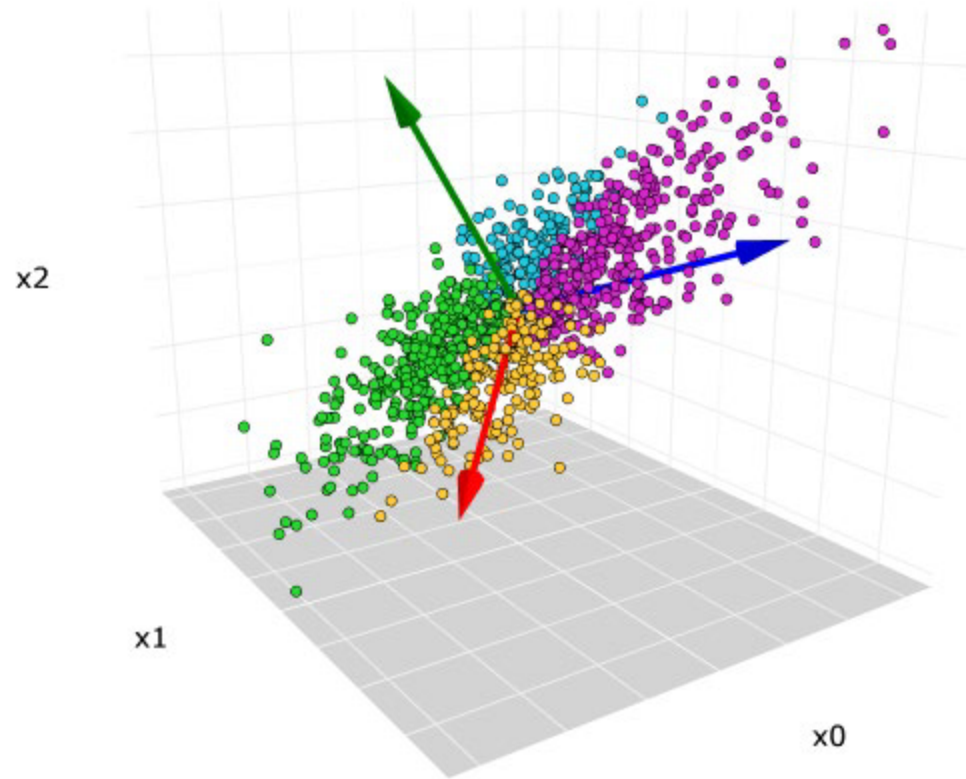
$$= \mathbf{E}(\mathbf{E}^\top \mathbf{D} \mathbf{E})\mathbf{E}^\top$$

$$= \mathbf{D}$$

- $\mathbf{P}$ should be the Eigenvector of $\frac{1}{n}\mathbf{X}\mathbf{X}^\top = \mathbf{C}_X$.
- The i^{th} diagonal value of $\mathbf{C}_Y$ is the variance of X along $\mathbf{p}_i$

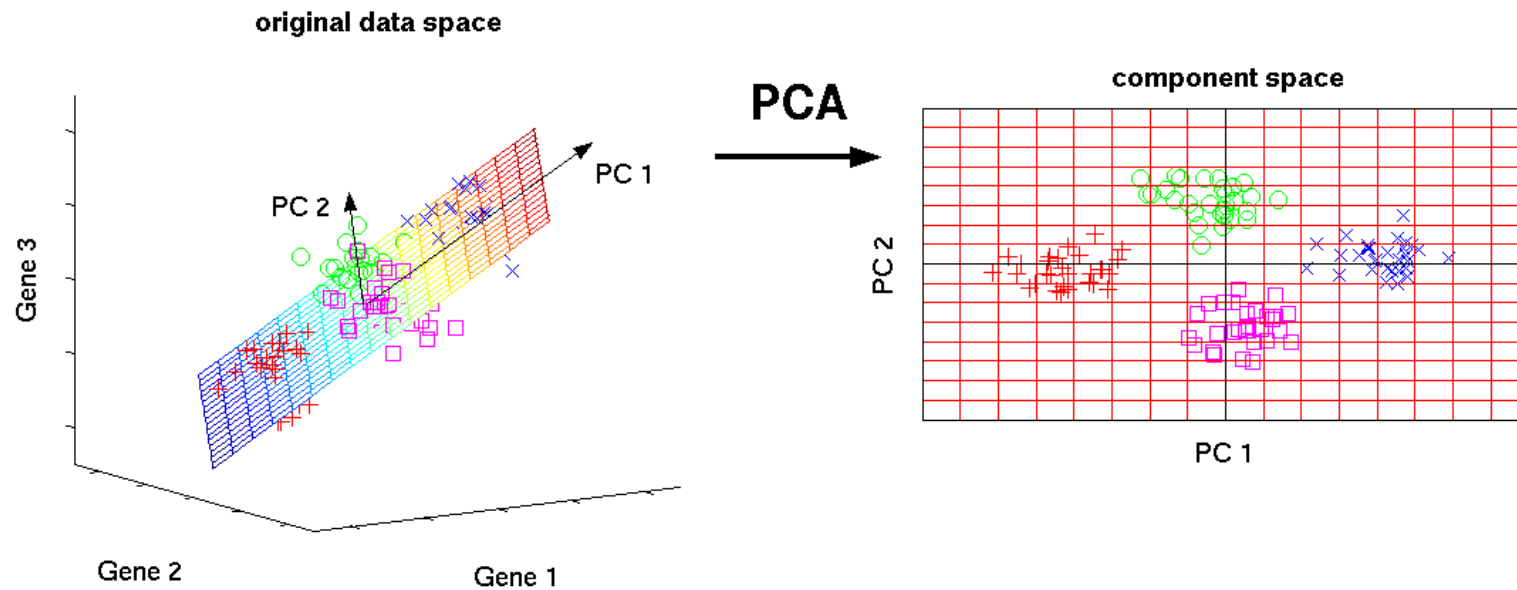
Visualized Example (1/2)

The covariance is maximized when data is projected on those three vectors.



3-D data have up-to 3 principal components.

Visualized Example (2/2)



You want to reduce the data dimension to 2-D, then choose the most important two principal components and project your data on them.

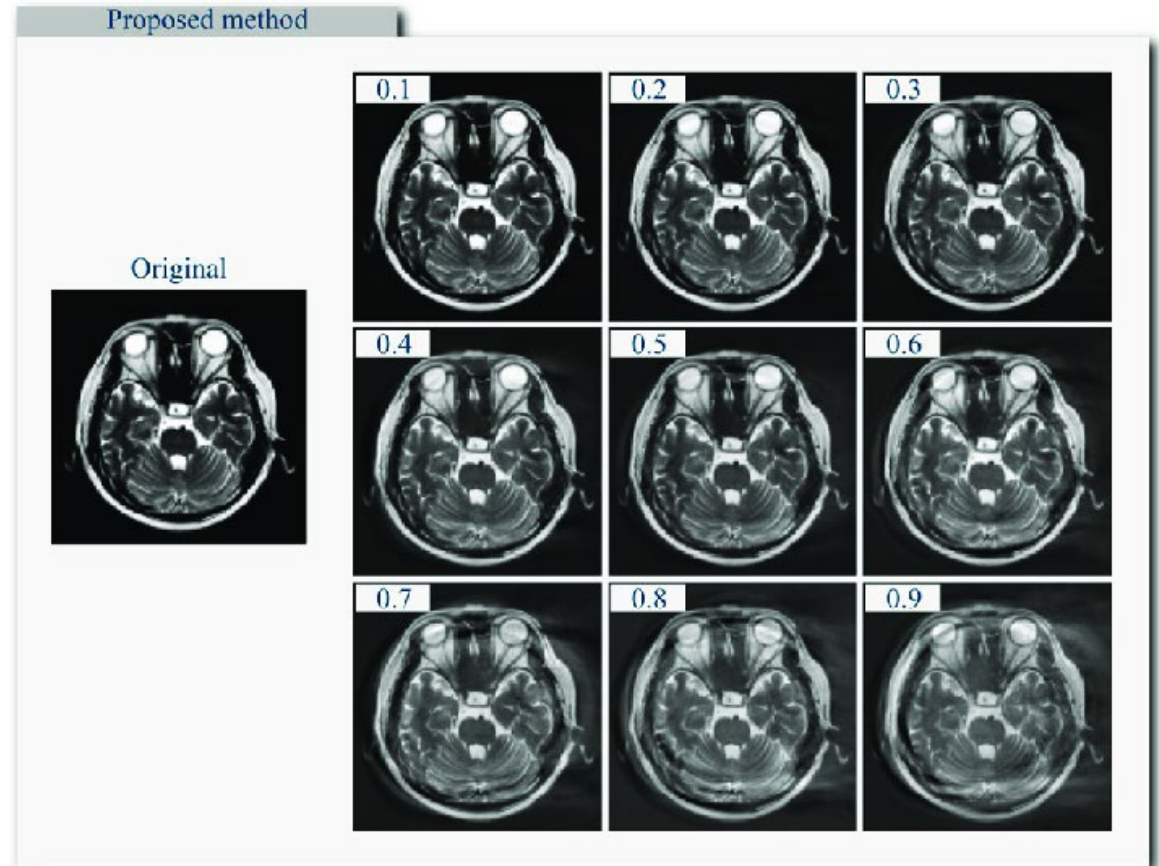
Advantages and Disadvantages

- ↑ • Removes noise from data.
- ↑ • Reduces data dimensionality.
- ↑ • May improve data visualization.
- ↓ • Sensitive to the scaling of variables.
- ↓ • Strong assumptions 대부분의 원천 데이터는 Orthogonal하지 않음
- ↓ • Computationally expensive. 고유값 분해의 계산 비용 매우 큼

Real-world Use-Cases (1/3)

- Medical Image Processing
 - Reduce the data dimensionality until the required information is revealed.
 - Noise may be removed as the data is represented in a lower-dimensional space.

대부분 환자들이 정보제공을 동의하지 않고 법적 규제가 심하므로 데이터 개수 적음, 그래서 PCA 사용 가능한 것



Real-world Use-Cases (2/3)

- Face Recognition
 - Together with the classification method (e.g, SVM) we learned earlier, PCA enables us to recognize individuals!



Real-world Use-Cases (3/3)

- Many features are available, but not all of them are informative!
 - PCA enables to find more important dimensions and find them to find inherent patterns in the observed data.

Index	Alcohol	Malic_Acid	Ash	Ash_Alcanity	Magnesium	Total_Phenols	Flavanoids	Nonflavanoid_Phenols	Proanthocyanins	Color_Intensity	Hue	OD280	Proline	Customer_Segment
0	14.23	1.71	2.43	15.6	127	2.8	3.06	0.28	2.29	5.64	1.04	3.92	1065	1
1	13.2	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.4	1050	1
2	13.16	2.36	2.67	18.6	101	2.8	3.24	0.3	2.81	5.68	1.03	3.17	1185	1
3	14.37	1.95	2.5	16.8	113	3.85	3.49	0.24	2.18	7.8	0.86	3.45	1480	1
4	13.24	2.59	2.87	21	118	2.8	2.69	0.39	1.82	4.32	1.04	2.93	735	1
5	14.2	1.76	2.45	15.2	112	3.27	3.39	0.34	1.97	6.75	1.05	2.85	1450	1
6	14.39	1.87	2.45	14.6	96	2.5	2.52	0.3	1.98	5.25	1.02	3.58	1290	1
7	14.06	2.15	2.61	17.6	121	2.6	2.51	0.31	1.25	5.05	1.06	3.58	1295	1
8	14.83	1.64	2.17	14	97	2.8	2.98	0.29	1.98	5.2	1.08	2.85	1045	1
9	13.86	1.35	2.27	16	98	2.98	3.15	0.22	1.85	7.22	1.01	3.55	1045	1
10	14.1	2.16	2.3	18	105	2.95	3.32	0.22	2.38	5.75	1.25	3.17	1510	1
11	14.12	1.48	2.32	16.8	95	2.2	2.43	0.26	1.57	5	1.17	2.82	1280	1
12	13.75	1.73	2.41	16	89	2.6	2.76	0.29	1.81	5.6	1.15	2.9	1320	1
13	14.75	1.73	2.39	11.4	91	3.1	3.69	0.43	2.81	5.4	1.25	2.73	1150	1
14	14.38	1.87	2.38	12	102	3.3	3.64	0.29	2.96	7.5	1.2	3	1547	1
15	13.63	1.81	2.7	17.2	112	2.85	2.91	0.3	1.46	7.3	1.28	2.88	1310	1

Wine Recommender System



Any Questions?

PCA is a useful statistical ML tool!

